

Groundwater quality evaluation using water quality index and geospatial techniques in parts of Anantapur District, Andhra Pradesh, South India

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ABSTRACT

The main aim of this research is to evaluate the quality of groundwater in parts of Anantapur district, Andhra Pradesh, for drinking and irrigation purposes. Groundwater samples were collected from various locations in the study area during the post-monsoon season and were analyzed for different parameters including pH, EC, TDS, Ca^{2+} , Mg^{2+} , Na^+ , K^+ , HCO_3^- , Cl^- , SO_4^{2-} , and NO_3^- . The results were then compared with the water quality standards of the World Health Organization (WHO). The major ions were found in the following order: $\text{Na}^+ > \text{Ca}^{2+} > \text{Mg}^{2+} > \text{K}^+ = \text{HCO}_3^- > \text{SO}_4^{2-} > \text{Cl}^- > \text{NO}_3^-$. The obtained data was then mapped using the Arc-GIS technique to show the spatial distribution pattern of physicochemical properties and major ion concentrations in groundwater. The water quality index showed that 69% of the samples are very poor, and 31% are unfit for drinking purposes due to effective leaching of ions, over-exploitation of groundwater, direct discharge of effluents, and agricultural impact. Based on the Gibbs diagram, the groundwater samples fall within the rock dominance region. The majority of the samples fell into the C3S1 category, which denotes water with high salinity and low sodium which is moderately suitable for irrigation purposes, according to the USSL diagram. According to the Wilcox classification, 27% and 2% of groundwater samples are deemed doubtful and unsuitable, respectively, while 69% of samples are deemed suitable for farming. In addition, most groundwater samples show moderate suitability for irrigation when evaluated for SAR, %Na, PI, RSC, MAR, and KR. Most of the samples, with the exception of a few spots, turned out to be suitable for irrigation. However, due to human activity, some samples had levels of contamination that were higher than allowed and needed to be treated before consumption.

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1. Introduction

In many regions of the world, the demand for groundwater has recently increased dramatically as a result of population growth, urbanization, industrialization, and intensive farming practices. Because there are few surface water supplies, the majority of Indian locations rely on groundwater resources for domestic, industrial, and agricultural uses, including drinking water (Adimalla et al., 2020a; Pasham et al., 2022; Chinthala et al., 2023). Therefore, it can be said that all countries recognize that having access to clean drinking water is both a fundamental human right and a requirement for human growth, health, and well-being (WHO, 2012). Water nowadays defines human, social, and economic development. It has also been seen as an endless and abundant resource. A number of significant environmental issues have arisen as a result of the alarming rate of population expansion, the development of industrial society, technological advancements, and the current trend of groundwater resource depletion. For many decades, there

have been growing concerns about the healthfulness of drinking water (Gowd, 2005; Aly et al., 2015; Papazotos et al., 2019; Agarwal et al., 2020). Therefore, understanding the hydrochemistry of groundwater is crucial to evaluating the excellence of groundwater, particularly in rural regions, which affects its suitability for household, irrigational, and industrial requirements. Because of this, groundwater is a particularly valuable natural resource. Groundwater-related environmental issues vary in severity from location to location based on the local geology, hydrology, weather conditions, and geochemical considerations (Lakshmi et al., 2021; Anusha et al., 2022; Kumar et al., 2022). However, investigations on the quality of groundwater for irrigation and drinking reasons have been done in a number of different places, including Uttar Pradesh (Ram et al., 2021; Etikala et al., 2021), Tamil Nadu (Balamurugan et al., 2020) and Maharashtra (Pande et al., 2020). In India, groundwater supplies >45% of all agricultural water and over 80% of home water needs, irrigating 39 million hectares (Gnanachandrasamy et al., 2015; Prasad et al., 2019; Golla et al., 2021). Additionally, India's most abundant and effective source of water for irrigation is groundwater (Krishnakumar et al., 2014; Jatwa et al., 2019). The main source of water for the region's household, potable, and farming

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needs is groundwater. Groundwater chemistry is a technique for raising the quality of the water for a variety of applications. Water quality evaluation techniques are classified into two types: WQI approaches and conventional methods, which compare measured parameter values to suggested levels (Deepa and Venkateswaran, 2018). In this regard, the quality of groundwater has been evaluated using a variety of approaches. One of these instruments is the WQI. This parameter is one of the best resources for obtaining a complete picture of water quality since it is derived from the chemical components of water. Furthermore, this parameter is a computational method that reduces a significant amount of data on water characteristics into a single value that represents overall water quality and is simple to understand for decision-makers and the general public (Etikala et al., 2019; Khatri et al., 2020). The geochemistry of groundwater and its appropriateness for potable and agricultural use in parts of the Anantapur district have yet to be thoroughly investigated. As a result, a detailed groundwater quality investigation and categorization of the subsurface was performed in order to determine its appropriateness for a variety of purposes, including domestic/drinking water and agriculture.

Numerous studies have been conducted in India and other parts of the world to evaluate the geochemical characteristics of groundwater quality for drinking and irrigation purposes (Krishan et al., 2016; Acharya et al., 2018; Adimalla et al., 2020b; Khatri et al., 2020; Mahmud et al., 2020). Researchers have utilized various classifications like electrical conductivity (EC), sodium absorption ratio (SAR), percent sodium (%Na), residual sodium carbonate (RSC), permeability index (PI), Kelley's ratio (KR), and magnesium adsorption ratio (MAR) for irrigation water quality assessment in different parts of the country (Safari et al., 2021; Giri et al., 2022; Samtio et al., 2023; Ravindra et al., 2023). Anantapur district is a hard rock terrain with no sufficient surface water, and thus, the majority of the district's population relies on groundwater for their daily needs. Bukkarayasamudram region in Anantapur district has witnessed a significant increase in groundwater usage, and various water quality issues have been observed in different locations. Therefore, it has been selected to evaluate the groundwater water quality issues with the primary objective of assessing groundwater quality for drinking and irrigation purposes.

Recent decades have seen a repeat of drought spells in these semi-arid regions, with excessive groundwater extraction contributing to the qualitative and quantitative deterioration of this vital resource. However, physical-chemical investigation and evaluation of the area's groundwater quality have elevated to a primary concern (Kumari and Rai, 2020). The spatial evolution of groundwater quality within the semi-arid regions has been assessed, based on a hydrogeochemical, WQI, and geospatial approach, to aid policymakers in their task of intervening for better management and protection of this resource.

2. Study area

The study region is situated between 14°35'00" and 14°50'00" N latitude and 77°35'00" and 77°45' 00" E longitude in parts of Anantapur district, Andhra Pradesh, India (Fig. 1), which covers an area of about 249 sq. km. The study area falls in the Survey of India toposheets 57F/10 and 57F/11. Anantapur is one of the country's most often drought-affected districts, with a mean annual precipitation of around 520 mm. During the summer months of March to May, temperatures in Anantapur range from 24° C to 46° C (Kumar et al., 2023). Agriculture provides a living for the majority of people in Anantapur, which has a mean elevation of 516 m. The lithological composition of the research area includes hornblende biotite gneiss, hornblende gneiss (HG), biotite gneiss (BG), migmatite, and grey or pink granites (Fig. 2).

3. Methodology

During the post-monsoon season, a total of 26 groundwater samples were collected from bore wells. Each sample was collected using

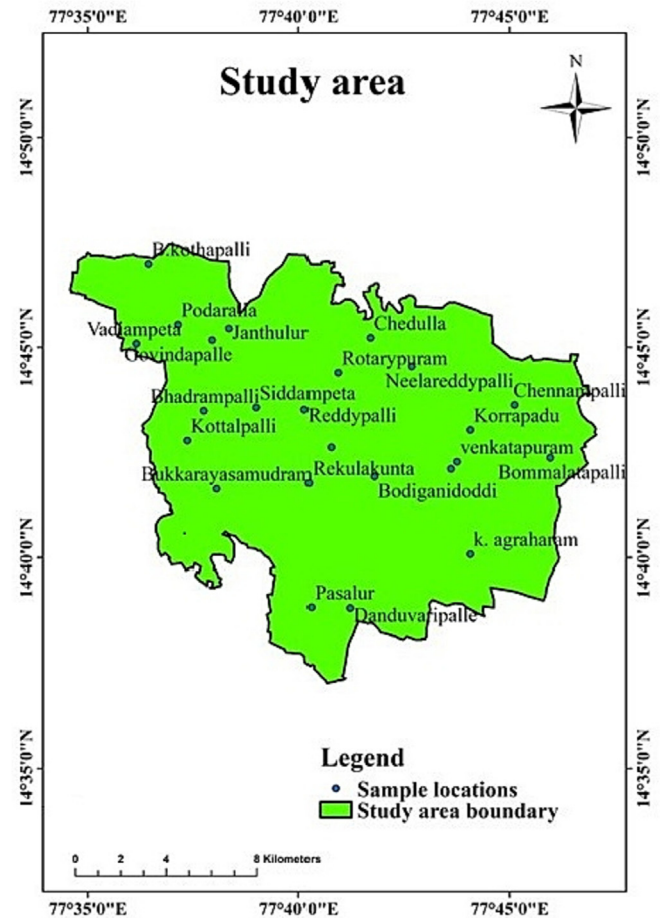


Fig. 1. Sample location map of the study area.

a one-liter acid-washed polyethylene bottle and measured for pH, EC, and TDS in the field using specialized equipment. The samples were carefully filled to avoid air bubbles and sealed with twin plastic lids to prevent moisture. To ensure their integrity, the samples were transported with special precautions to prevent movement and kept at room temperature. At the laboratory, concentrations of Na^+ and K^+ were determined using a flame photometer, while volumetric methods were used to assess total hardness as CaCO_3 , Ca^{2+} , CO_3^{2-} , HCO_3^- , and Cl^- . Mg^{2+} was determined using total hardness and calcium, while spectrophotometry was used to determine SO_4^{2-} and NO_3^- .

All groundwater samples were analyzed for the ionic balance error (IBE) among total cations and total anions. The IBE was found to be within the allowed tolerance of $\pm 5\%$.

$$\text{IBE} = \frac{\sum \text{Cations} - \sum \text{Anions}}{\sum \text{Cations} + \sum \text{Anions}} \times 100 \quad (1)$$

3.1. GIS analysis

The basic map for the research area was created using the Survey of India topographic sheets 57F/10 and 57F/11 through ArcGIS 10.8 program (Fig. 1). Garmin GPS was utilized to locate each sampling site, and the coordinates were then entered into a GIS platform for further research. To interpolate values between observations, the inverse distance weighted interpolation (IDW) technique was employed (Kumar et al., 2021; Kumar et al., 2022b; Badapalli et al., 2022). The IDW method calculates weights using the inverse distance method from the

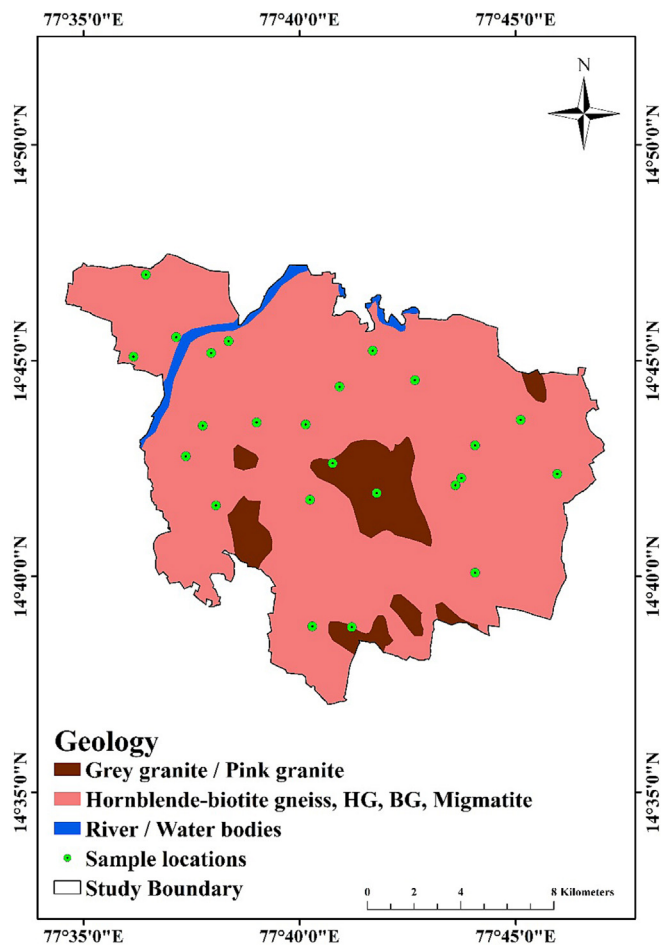


Fig. 2. Geological map of the study area.

Table 1
Physico-chemical parameters of groundwater of the study area.

S. No	Sample locations	pH	EC	TDS	Ca ²⁺	Mg ²⁺	Na ⁺	K ⁺	HCO ₃ ⁻	Cl ⁻	SO ₄ ²⁻	NO ₃ ⁻	TH
1	Bodiganidoddi	7.8	910	480	89	14	126	5	528	124	170	10	280
2	Bukkarayasamudram pond	7.6	2600	1450	136	21	238	8	485	124	198	15	426
3	Bukkarayasamudram	7.5	2360	1210	128	18	217	15	458	106	182	0	394
4	Reddypalli	7.8	1770	890	164	74	112	5	448	106	411	9	713
5	Korrapadu	8.2	1870	960	137	18	144	6	427	89	271	0	416
6	Neelareddypalli	8.3	1690	860	82	16	108	2	552	142	59	0	271
7	Chennampalli	7.7	890	450	95	34	156	3	427	124	93	0	377
8	Chennarayunipalli	7.6	1480	780	94	18	128	3	462	195	129	0	309
9	Danduvaripalle	7.4	1260	640	148	21	98	14	514	124	157	0	456
10	Janthulur	7.9	1080	580	145	66	89	6	648	213	87	0	633
11	Podaralla	7.5	1890	960	129	10	104	8	722	248	59	0	364
12	Siddampeta	7.4	1150	580	113	44	96	4	481	160	126	0	463
13	B. Kothapalli	8.1	2470	1250	80	42	112	6	432	142	49	0	372
14	Dayyalakuntapalli	7.8	3620	1820	185	22	92	11	526	213	37	22	553
15	Boya kottala	7.9	2680	1420	148	58	86	14	574	198	48	14	608
16	Vadiampeta	7.5	2240	1130	131	100	108	14	446	266	23	5	738
17	Venkatapuram	8.1	1470	810	186	106	132	7	418	160	107	34	900
18	Bommalatapalli	8.2	1750	880	152	105	116	17	592	142	24	0	811
19	Pasalur	8.1	1620	820	195	135	97	22	462	151	33	0	1041
20	Rotary Puram	7.9	2010	1020	198	59	114	7	628	168	46	25	737
21	Govindapalle	7.6	1320	680	124	92	82	3	658	177	102	0	687
22	Chedulla	7.8	1580	810	129	118	164	38	532	213	154	2	806
23	K. Agraharam	7.7	1640	840	193	138	172	4	782	213	128	0	1048
24	Rekulakunta	7.6	1170	610	128	78	129	5	628	168	64	0	640
25	Bhadrapalli	7.4	960	510	96	46	142	4	532	239	79	0	429
26	Kottalpalli	7.9	2680	1360	50	31	117	2	423	195	54	0	252
Min		7.4	890	450	50	10	82	2	418	89	23	0	252
Max		8.3	3620	1820	198	138	238	38	782	266	411	34	1049
Avg		7.8	1775	915	133	57	126	9	530	170	111	5.2	566
St. dev		0.3	663.9	340.4	39.1	40.4	38.1	7.9	98.8	47.2	86.8	9.3	233
WHO (2012)		6.5–8.5	—	500	75	30	200	—	—	250	250	45	—

observation location to the estimation point, resulting in each value representing the weighted average of neighboring locations. The IDW technique, when extended through spatial analyst software extension, is an invaluable tool for creating spatial variation maps of the subsurface physicochemical properties of these locations. The IDW technique has been widely used across the globe for a variety of reasons, producing concentration zone-distinguishing spatial distribution maps.

3.2. Evaluation of groundwater quality using the water quality index (WQI)

The Water Quality Index is a one-of-a-kind quantification that delineates the main reasons for inconsistencies in water quality (Singha and Pasupuleti, 2020; Krupavathi et al., 2022). The WQI is calculated using elements of water suitability for people. WQI was calculated using pH, electrical conductivity, chloride, dissolved solids, hardness, sulfate, sodium, manganese, calcium, potassium, and magnesium (Chaurasia et al., 2018). The WQI is a weighted calculation utilized in various studies to make these kinds of conclusions.

3.3. Weighted arithmetic water quality index

By combining the most widely evaluated water quality parameters, the weighted arithmetic water quality index explains water quality according to its level of excellence. It's been utilized by a lot of scientists, including:

$$K = \frac{1}{\sum (\frac{1}{Si})}$$
 (2)

where K represents proportionality and is fairly constant, and Si is a parameter standard. The optimal (ideal) value of the parameter was used to calculate the sub-index (Wi):

$$Wi = \frac{K}{Si}$$
 (3)

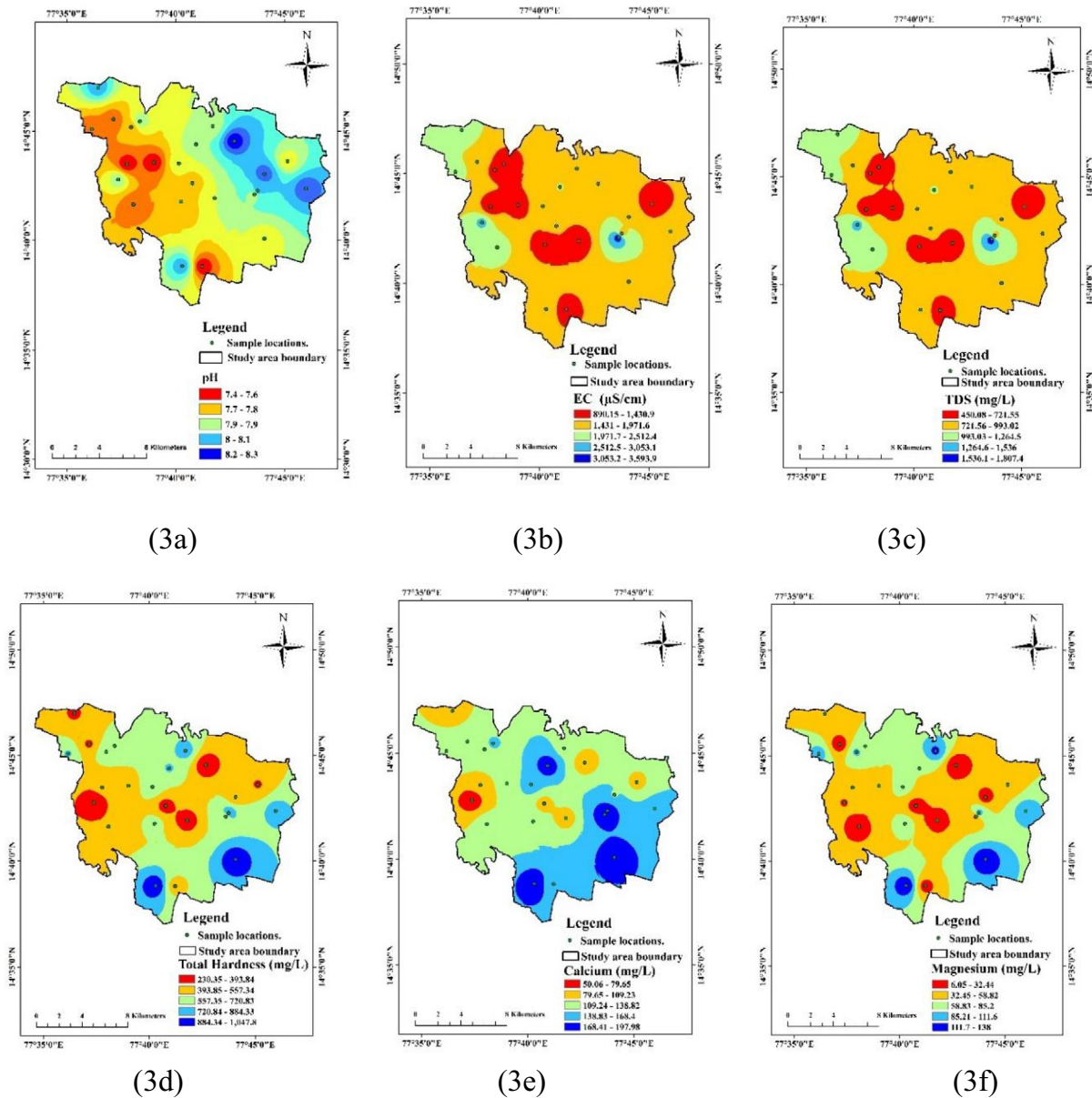


Fig. 3. Spatial distribution maps of pH (a), EC (b), TDS (c), TH (d), Ca^{2+} (e), Mg^{2+} (f), Na^+ (g), K^+ (h), HCO_3^- (i), Cl^- (j), SO_4^{2-} (k), NO_3^- (l).

The sub-index was calculated using the procedure below (Qi).

$$Q_i = 100 \left[\frac{V_i - V_o}{S_i - V_o} \right] \quad (4)$$

where v_i represents the parameter content in water analysis, S_i represents the parameter's standard accurate value, and V_o represents the parameter's true value in fresh water.

$$WQI = \frac{\sum W_i Q_i}{\sum W_i} \quad (5)$$

where WQI is a number between 0 and 100 that represents the quality of drinking water; Q_i is a relative measure of water specific to each parameter; and i is the parameter range.

4. Results and discussion

In Table 1, we have summarized the physicochemical parameters that were determined and compared them to the standards of WHO, 2012.

4.1. pH

The pH of water is the most significant parameter and is valuable in estimating the overall quality of water. The pH level of the groundwater in the study area ranged from 7.4 to 8.3 with an average value of 7.8, indicating a strong alkaline predominance. All the water samples in the study area are within the recommended levels of 6.5–8.5 (WHO, 2012). Fig. 3a shows the distribution pattern of pH.

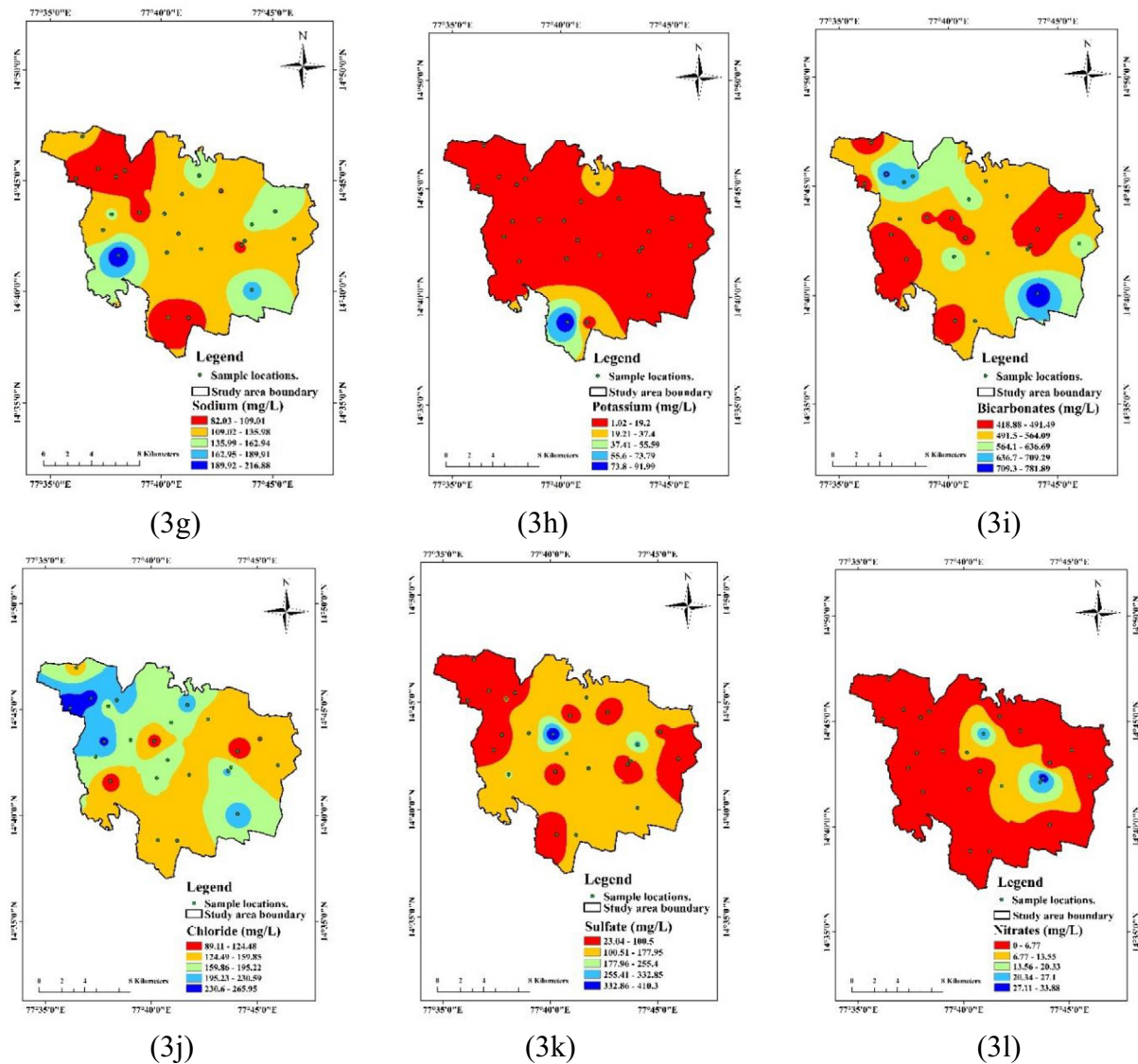


Fig. 3 (continued).

4.2. Electrical conductivity (EC)

Electrical conductivity is a measure of ion concentrations that depends on the temperature and type of ions and their concentrations in the water. The observed EC levels ranged from 890 $\mu\text{S}/\text{cm}$ to 3620 $\mu\text{S}/\text{cm}$, and all the samples exceeded the acceptable limit set by WHO, 2012. High EC levels in groundwater suggest a higher concentration of dissolved salt content and greater attentiveness level. This also indicates the existence of multiple aquifer systems and regional differences in soil types. Fig. 3b shows the distribution pattern of EC.

4.3. Total dissolved solids (TDS)

TDS refers to the collective amount of soluble inorganic and organic ions present in water. TDS levels in the research region range from 450 to 1820 mg/L, with an average of 915.4 mg/L. Drinking water should contain no >500 ppm of total dissolved solids (WHO, 2012). The highest useful level for TDS was exceeded in several locations, making the water unfit for drinking. Based on groundwater categorization, almost 75% of

the groundwater sources were freshwater (as shown in Table 2) and were safe to drink, while the other 25% were not. The high TDS levels were due to the disposal of industrial and municipal wastes, commercial leaks, and the infiltration of solids-laden channel water. Fig. 3c shows the distribution pattern of TDS.

4.4. Total hardness (TH)

Total hardness is determined by the quantity of magnesium and calcium-containing suspended particulates in the water. The research region's total hardness ranges from 252 to 1049 mg/L, with an average of 566 mg/L (Table 1). The findings indicated that the hardness of most samples exceeded the permitted limit set by WHO, 2012 guidelines. Hard water is not recommended for home usage because it can cause metal dissolution, leading to abrasive coatings under pipelines, heaters, and drums. It may also lower the observed quality of water, putting human health at risk by producing disorders like kidney diseases, microcephaly, prenatal mortality, leukemia, and cardiovascular problems. Fig. 3d shows the distribution pattern of TH.

Table 2
Calculation of WQI for individual water samples.

S. No	Sample locations	Latitudes	Longitudes	WQI	Water quality status
1	Bodiganidoddi	14° 44' 16.87"N	77°40'51.22"E	77.73	Very poor water
2	Bukkarayasamudram Pond	14° 41' 15.72" N	77° 41' 47.4" E	95.69	Very poor water
3	Bukkarayasamudram	14° 41' 37.55"N	77° 37'59.38"E	83.15	Very poor water
4	Reddypalli	14° 43' 6.71"N	77° 40'40.27"E	100.04	Unsuitable for drinking
5	Korrapadu	14° 45' 11.32"N	77° 42'39.36"E	87.07	Very poor water
6	Neelareddypalli	14° 44' 32.64"N	77° 42'40.32"E	77.8	Very poor water
7	Chennampalli	14° 43' 37.56"N	77° 45' 7.2"E	78.79	Very poor water
8	Chennarayunipalli	14° 42' 37.08"N	77° 40'45.84"E	77.03	Very poor water
9	Danduvaripalle	14° 38' 49.2"N	77° 41'12.48"E	80.1	Very poor water
10	Janthulur	14° 45' 27" N	77° 38 21.48"E	92.84	Very poor water
11	Podaralla	14° 45' 32.4"N	77° 37' 8.76"E	79.29	Very poor water
12	Siddampeta	14° 43' 33.96"N	77° 39' 0.36"E	81.24	Very poor water
13	B. Kothapalli	14° 46' 59.16"N	77° 36' 26.64"E	86.03	Very poor water
14	Dayyalakuntapalli	14° 42' 6.48"N	77° 43'36.48"E	99.32	Very poor water
15	Boya Kottala	14° 42' 6.84" N	77° 43' 36.84" E	87.47	Very poor water
16	Vadiampeta	14° 43' 27.48"N	77° 37' 48.72"E	100.46	Unsuitable for drinking
17	Venkatapuram	14° 42' 16.56"N	77° 43' 44.76"E	115.14	Unsuitable for drinking
18	Bommalatapalli	14° 42' 22.32"N	77° 45' 57.96"E	105.15	Unsuitable for drinking
19	Pasalur	14° 38' 50.28"N	77° 40' 17.76"E	114.27	Unsuitable for drinking
20	Rotarypuram	14° 44' 23.64"N	77° 40' 55.56"E	105.22	Unsuitable for drinking
21	Govindapalle	14° 44' 18.6"N	77° 36' 45.72"E	94.71	Very poor water
22	Chedulla	14° 45' 13.68"N	77° 41' 41.64"E	105.03	Unsuitable for drinking
23	K. Agharam	14° 40' 4.8"N	77° 44' 3.84"E	114.87	Unsuitable for drinking
24	Rekulakunta	14° 41' 46.32"N	77° 40' 14.52"E	91.79	Very poor water
25	Bhadrapalli	14° 43' 30.36"N	77° 37' 46.56"E	80.45	Very poor water
26	Kottalpalli	14° 39' 28.91"N	77° 39' 22.71"	80.53	Very poor water

4.5. Major cations and anions

The distribution of major ions in the groundwater is found in the following order: $\text{Na}^+ > \text{Ca}^{2+} > \text{Mg}^{2+} > \text{K}^+$ and $\text{HCO}_3^- > \text{SO}_4^{2-} > \text{Cl}^- > \text{NO}_3^-$. The concentration of calcium in water is primarily influenced by the presence of carbonates, sulfate, and chlorides. The hardness of the water is determined by the amount of calcium and magnesium salts present. The acceptable range of calcium concentration in drinking water typically falls between 75 and 200 mg/L. Calcium levels in the study region range from 50 to 198 mg/L, with almost 96% of samples exceeding the permissible amount of 75 mg/L. However, all samples were within the permitted limit of 200 mg/L (WHO, 2012). Magnesium (Mg^{2+}) is a common element found in the Earth's crust. Its hydrochemistry is similar to calcium, but its solubility can vary based on the presence of carbon dioxide. Ferromagnesium minerals and waste drainage are the primary sources of magnesium in natural water. The World Health Organization (WHO, 2012) has set the recommended limit for magnesium in drinking water between 30 and 150 mg/L. This limit is established to prevent excessive magnesium intake, which can lead to several health risks such as diarrhea, nausea, and irregular heartbeat. Therefore, monitoring the magnesium content in drinking water is crucial to ensure its safety and quality for human consumption. Mg^{2+} content in the research region varied from 10 to 138 mg/L, with a mean of 58 mg/L. Sodium is an abundant element that is widely found in soils, plants, and water on Earth. Due to the high solubility of sodium salts, sodium ions are present almost everywhere in water. Sodium levels in the study region range from 82 to 238 mg/L, with a mean of 126 mg/L. According to the WHO, 2012 guidelines, the maximum permissible limit is 200 mg/L. Nearly 4% of

Table 3
The WQI range and type of water can be classified below: (Brown et al., 1972).

WQI value	Class	Water quality status	Percentage of water samples
>25	I	Excellent	—
26–50	II	Good	—
51–75	III	Poor	—
76–100	IV	very poor	69%
>100	V	Unsuitable for drinking	31%

samples exceeded the permissible limits for sodium concentration, which can cause health issues such as high blood pressure, heart problems, brain diseases, and kidney failure. Potassium levels in water

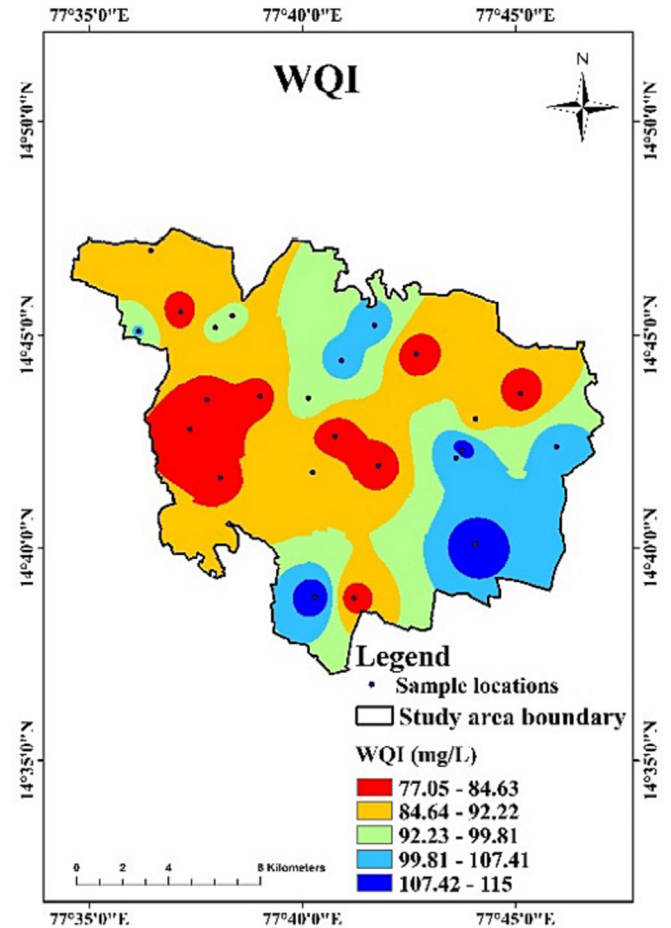


Fig. 4. Spatial distribution map of WQI.

that exceed a few tens of parts per million (ppm) indicate contamination. Potassium has the lowest concentration in groundwater of any cation. According to 2012 WHO recommendations, a potassium value of <10 ppm is deemed suitable for drinking water. Potassium levels in the study region range from 2.0 to 38 mg/L, with a mean of 9.0 mg/L. Alkalinity, which is the capacity of water to resist changes in pH, is determined by the amount of carbonate, bicarbonate, and hydroxide ions present in the water. In the research area, the bicarbonate concentration of the water samples ranged from 418 to 782 mg/L with a mean of 530 mg/L. Bicarbonate ions in the water can help neutralize any acidic substances present in the water, making it less acidic and more balanced in terms of pH. Chloride is a mineral

that occurs naturally in water and is also found in salt and sometimes in combination with potassium and calcium. Cl^- levels in the study region range from 89 mg/L to 266 mg/L, with a mean of 170 mg/L. The chloride level was higher than the recommended limit of 250 mg/L in 82.14% of samples, indicating a salty taste in the groundwater. Sulfate is a significant dissolved component found in rainwater, and it is also present in some soils and rocks in the form of sulfate minerals. The World Health Organization (WHO, 2012) recommends that the sulfate levels in drinking water should be between 250 and 400 mg/L. Maintaining this recommended range is essential for ensuring the water quality is safe for human consumption. High levels of sulfate in drinking water can cause gastrointestinal issues, while low levels may result in

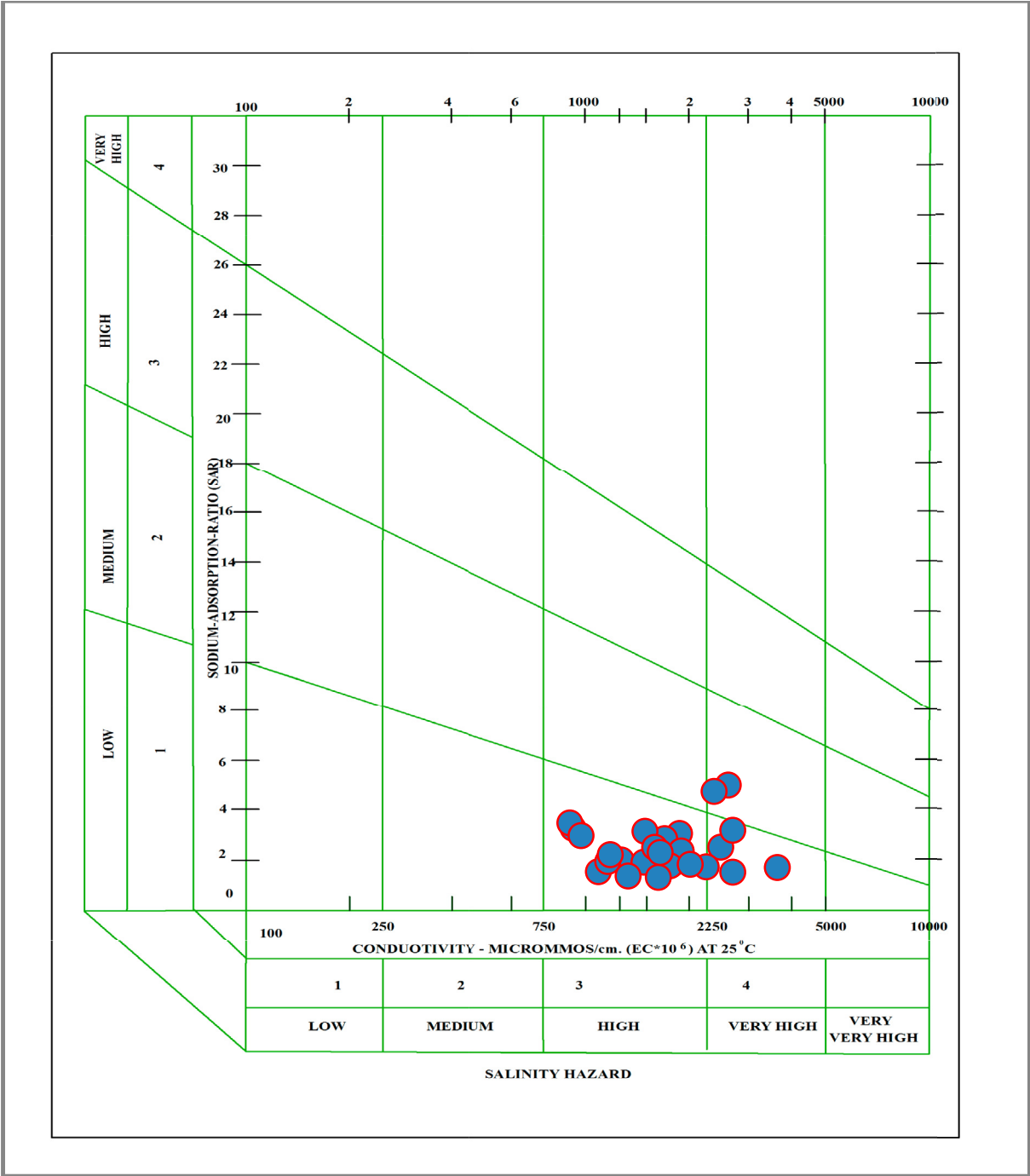


Fig. 5. Irrigation quality of groundwater (USSL, 1954).

the water being unsuitable for industrial use. SO_4^{2-} concentrations ranged from 23 to 411 mg/L, with a mean of 111 mg/L. Nitrates are essential nutrients that provide nitrogen, a vital element for the growth and development of plants and other vegetation. However, excessive amounts of nitrates in the environment can cause serious problems, such as nitrate contamination of groundwater. This contamination is often caused by anthropogenic activities, including the use of fertilizers and the disposal of animal waste. The nitrate content varied from 0 to 34 mg/L, with a mean of 5.2 mg/L. High levels of nitrate in water can lead to respiratory problems in children and health issues such as stomach cancer, thyroid disease, and high blood pressure in adults. Fig. 3e, f, g, h, i, j, k, l illustrate the spatial distribution of all cations (Ca^{2+} , Mg^{2+} , Na^+ , K^+) and anions (HCO_3^- , Cl^- , SO_4^{2-} , NO_3^-).

4.6. Water quality index (WQI)

WQI at individual sampling locations is shown in Table 2. The computed WQI values in this study vary from 77.03 to 115.14. As a result, this can be divided into five categories: Excellent, Class I (25); Good, Class II (26–50); Poor, Class III (51–75); Very poor,

Class IV (76–100); Unsuitable for drinking, Class V (>100). According to the water quality index classification, 69% of the groundwater samples fall into Class IV: very poor, while the remaining 31% fall into Class V: unsuitable for drinking, as indicated in Table 3. The high value of WQI may be related to greater amounts of TDS, EC, and TH. These findings suggest that the groundwater is just slightly suitable for consumption. Fig. 4 shows a spatial distribution map of WQI.

4.7. Irrigation water quality

Higher ion concentrations in irrigation water reduce osmotic activity in plant tissues, which affects soil structure and, as a result, plant development. It prevents water from running off the leaves and stems, affecting farm productivity. Groundwater for irrigation is commonly assessed using salinity hazards, sodium hazards, %Na, PI, RSC, MAR, and Kelly's ratio (KR). Salinity hazards are caused by poor water conditions, resulting in high salinity and crop failure. Due to the interchange of Na^+ for Ca^{2+} in soils, sodium hazard renders soil compact and impermeable (Todd and Mays, 2004). As a result, it slows the growth of plants.

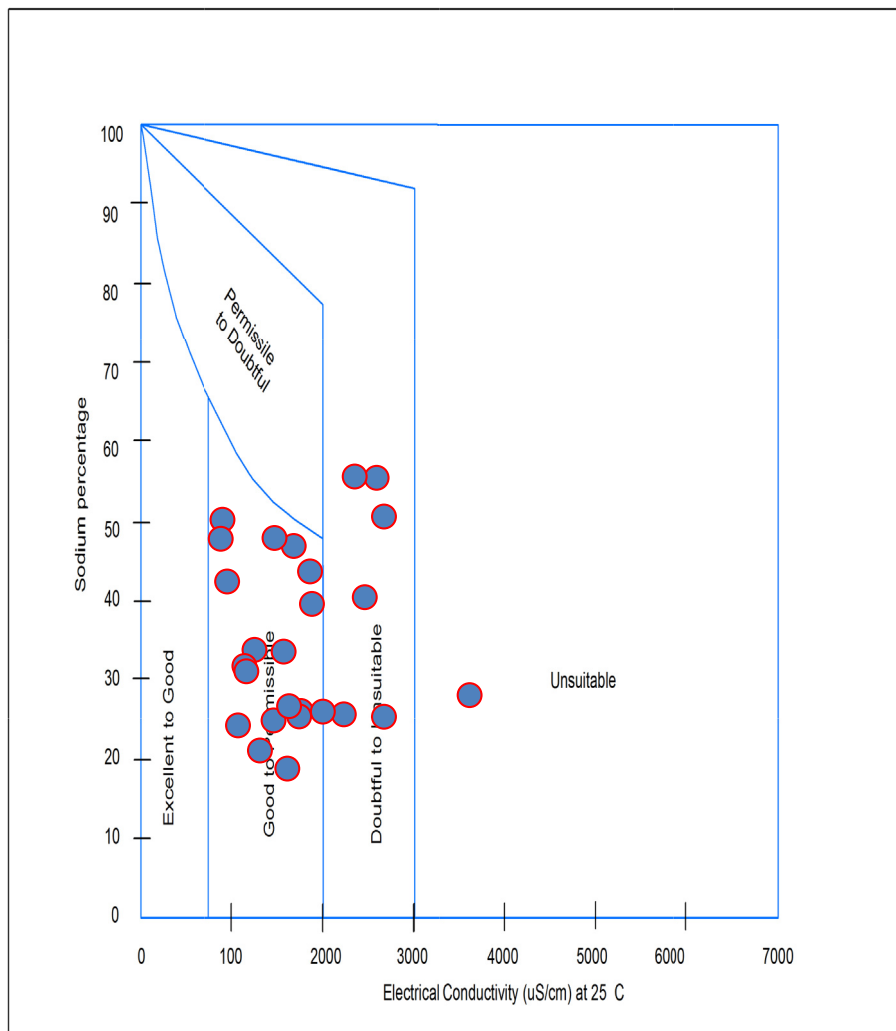


Fig. 6. Irrigation water quality of groundwater (After Wilcox, 1955).

4.8. Sodium adsorption ratio (SAR)

The EC ranges from 890 to 3620 $\mu\text{S}/\text{cm}$, with an average of 1703 $\mu\text{S}/\text{cm}$ (Table 1). The SAR ranges from 1.31 to 5.03, with an average of 2.47 (Table 5). USSL's diagram depicts the full influence of EC and SAR on plant development (Fig. 5). Most groundwater sources (about 77%) come from the C3S1 zone, which denotes high conductivity and low sodium toxicity water that can be used moderately for irrigation. The C4S1 zone is 15% away, indicating a high conductivity and low sodium threat 8% from the C4S2 zone, indicating very high conductivity and a medium sodium threat (Table 5). This suggests a progressive increase in the salinity and salt hazards associated with irrigation (Tiwari et al., 2017; Rezaei et al., 2020). Thus, semi-tolerant to salt-tolerant crops can be chosen using particular soil treatment procedures.

$$\text{SAR} = \frac{\text{Na}^+}{\sqrt{\frac{(\text{Ca}^{2+} + \text{Mg}^{2+})}{2}}} \tag{6}$$

4.9. Percent sodium (%Na)

The permeability of soil is reduced when sodium replaces calcium. The quality of irrigation water is classified using the %Na values provided by Wilcox (1955). Sodium combines with carbonates and chloride to produce alkaline and saline soils. Alkaline or saline soils are harmful to plant growth. Sodium content is frequently represented as a percentage of sodium (%Na), which is derived following Wilcox's (1955) formula (7):

$$\% \text{Na} = \frac{\text{Na}^+ + \text{K}^+}{(\text{Ca}^{2+} + \text{Mg}^{2+} + \text{Na}^+ + \text{K}^+)} \times 100 \tag{7}$$

In the study region, the %Na value ranged from 18.67 to 55.53, with a mean value of 35.37 respectively (Table 5). According to Wilcox's (1955) classification, 4%, 58%, and 38% of samples fall under the excellent, good, and permissible for irrigation (Table 5).

4.10. Wilcox diagram

Wilcox (1955) established a model for determining the effectiveness of irrigated agriculture based on the integration of EC and percent Na^+ (Fig. 6). Excellent to good, good to permissible, permissible to doubtful, doubtful to unsuitable, and inappropriate are the five zones in the diagram, with rising salinity and sodium threat for cultivation. 69% of groundwater sources are in the good to permitted zone, 27% are in the doubtful to inappropriate zone, and 4% are in the inappropriate zone for irrigation (Table 5).

4.11. Permeability index (PI)

Plant life is dependent on soil permeability. It should not help plant development if it is low. It is heavily influenced by soil Na^+ , Ca^{2+} , Mg^{2+} , and HCO_3^- levels, as well as long-term groundwater recharge with high salt concentration (Swarna Latha and Nageswara Rao, 2012; Subba Rao, 2018). Doneen (1964) described permeability in terms of the permeability index (PI), which is computed using Eq. (8), with all ions stated in meq/L :

$$\text{PI} = \frac{\text{Na}^+ + \sqrt{\text{HCO}_3^-}}{(\text{Ca}^{2+} + \text{Mg}^{2+} + \text{Na}^+)} \times 100 \tag{8}$$

Water quality is divided into three categories based on PI (Table 5). The class I category is good for irrigation; class II is slightly acceptable; and class III is not. In the research area, the PI ranges from 27.79 to 76.18 with a mean of 53.22 (Table 5). According to the PI categorization,

Table 4
Groundwater quality indices of irrigation water samples in the research area.

Sample No.	SAR (meq/L)	RSC (meq/L)	PI (meq/L)	KR (meq/L)	MAR (meq/L)	Indices of base exchange		Gibbs ratio I (meq/L)	Gibbs ratio II (meq/L)	%Na (meq/L)	Soltan classification	
						CAI- 1 (meq/L)	CAI- 2 (meq/L)				R1 (meq/L)	R2 (meq/L)
1	3.28	3.01	75.96	0.98	20.58	−0.6	−0.17	0.29	0.55	50.07	0.56	0.60
2	5.03	−0.61	69.77	1.22	20.28	−2.02	−0.59	0.31	0.6	55.36	1.66	1.71
3	4.77	−0.4	70.32	1.2	18.81	−2.29	−0.61	0.29	0.6	55.53	1.70	1.80
4	1.82	−6.96	39.56	0.34	42.64	−0.67	−0.13	0.29	0.37	25.95	0.22	0.24
5	3.07	−1.36	61.07	0.75	17.79	−1.56	−0.31	0.27	0.48	43.56	0.67	0.69
6	2.86	3.59	76.18	0.87	24.32	−0.19	−0.07	0.31	0.53	46.76	0.57	0.61
7	3.5	−0.58	65.82	0.9	37.09	−0.96	−0.38	0.33	0.59	47.66	1.70	1.74
8	3.16	1.36	70.78	0.9	23.98	−0.03	−0.01	0.42	0.54	47.78	0.03	0.05
9	2	−0.73	53.48	0.47	18.95	−0.32	−0.1	0.29	0.37	33.65	0.23	0.34
10	1.54	−2.1	43.08	0.31	42.85	0.33	0.16	0.36	0.35	24.12	−1.18	−1.10
11	2.37	4.51	67.5	0.62	11.32	0.32	0.17	0.37	0.41	39.44	−2.01	−1.85
12	1.94	−1.42	51.94	0.45	39.08	0.05	0.02	0.37	0.43	31.61	−0.13	−0.09
13	2.52	−0.4	61.07	0.65	46.38	−0.25	−0.13	0.36	0.55	40.3	0.85	1.00
14	1.7	−2.47	46.08	0.36	16.38	0.29	0.18	0.41	0.3	27.95	−2.60	−2.24
15	1.51	−2.8	42.79	0.31	39.23	0.27	0.14	0.37	0.34	25.22	−1.85	−1.49
16	1.73	−7.49	38.03	0.32	55.7	0.33	0.32	0.51	0.42	25.51	−5.86	−5.11
17	1.91	−11.18	35.19	0.32	48.42	−0.31	−0.16	0.4	0.38	24.76	0.55	0.63
18	1.77	−6.57	38.36	0.31	53.23	−0.37	−0.15	0.29	0.4	25.26	2.09	2.95
19	1.31	−13.3	27.79	0.2	53.28	−0.12	−0.06	0.36	0.3	18.67	−0.06	0.76
20	1.83	−4.49	41.44	0.34	32.92	−0.08	−0.04	0.32	0.33	25.86	0.23	0.42
21	1.36	−3.02	39.48	0.26	55	0.27	0.1	0.32	0.37	20.95	−0.67	−0.64
22	2.51	−7.47	43.29	0.44	60.11	−0.35	−0.18	0.41	0.53	33.43	0.35	0.65
23	2.31	−8.23	38.84	0.36	54.08	−0.26	−0.1	0.32	0.44	26.56	0.55	0.59
24	2.22	−2.56	47.86	0.44	50.1	−0.21	−0.09	0.32	0.47	30.96	0.66	0.75
25	2.98	0.1	61.82	0.72	44.11	0.07	0.04	0.44	0.56	42.28	−0.34	−0.28
26	3.2	1.85	76.18	1.01	50.53	0.07	0.04	0.44	0.67	50.48	−0.36	−0.32
Min	1.31	−13.30	27.79	0.20	11.32	−2.29	−0.61	0.27	0.30	18.67	−5.86	−5.11
Max	5.03	4.51	76.18	1.22	60.11	0.33	0.32	0.51	0.67	55.53	2.09	2.95
Avg	2.47	−2.68	53.22	0.58	37.58	−0.33	−0.08	0.35	0.46	35.37	−0.09	0.09

88% of groundwater samples are good for agriculture (class II), while the remaining 12% are excellent (class I) for irrigation (Table 5).

4.12. Residual sodium carbonate (RSC)

When there are more carbonates ($\text{HCO}_3^- + \text{CO}_3^{2-}$) than alkaline earths, they interact with Na^+ to form NaHCO_3 (Eq. (9)). It has an impact on the formation of the soil (Eaton, 1950a, 1950b). “Residual sodium carbonate” is the term for this (RSC; Eq. (10)). The lower the soil permeability and, as a result, crop production, the stronger the adsorption of Na^+ in the soil.



$$\text{RSC} = (\text{HCO}_3^- + \text{CO}_3^{2-}) + (\text{Ca}^{2+} + \text{Mg}^{2+}) \quad (10)$$

RSC values range from -13.30 to 4.51 , with a mean of -2.68 (Table 4). Because of the constant growth in Na^+ levels in soils, which results in a limited penetration rate, RSC assesses around 81% of groundwater samples as acceptable, 8% as somewhat suitable, and 11% as unfit for agriculture (Table 5).

4.13. Magnesium adsorption ratio (MAR)

Ca^{2+} and Mg^{2+} ions are commonly in balance in water. But they act differently in the soil system. Magnesium declines in soil structure when the water's salinity and sodium content are higher. In soil and water, exchangeable Na^+ causes a high amount of Mg^{2+} (Wang, 2013; Sarkar et al., 2022). More Mg^{2+} in equilibrium influences soil quality by making it alkaline. As a result, it has an impact on crop production. This impact on agricultural output is calculated using the magnesium ratio, as given in Eq. (11).

$$\text{MAR} = \frac{\text{Mg}^{2+}}{(\text{Ca}^{2+} + \text{Mg}^{2+})} \times 100 \quad (11)$$

If the MAR in water exceeds 50, it is regarded as unfit for irrigation due to its negative impact on agricultural output. As a result, such water is unfit for farming. The MAR in the research area has a range from 11.32 to 60.11 with a mean of 37.58 (Table 4). According to MAR categorization, 69% of groundwater samples are appropriate for irrigation, whereas 31% are inappropriate (Table 5), because magnesium disrupts soil structure, which reduces crop yield.

4.14. Chloralkaline indices (CAI)

Schoeller (1977) discovered that groundwater's chemical composition changes over time and that its movement can be described by the chloralkaline indices CAI I and CAI II (Eq. (12) and Eq. (13)). Na^+ , K^+ , and Ca^{2+} from the water and Mg^{2+} and Ca^{2+} from the rocks directly exchanged ions because of the presence of a positive chloralkaline guide. Mg^{2+} and Ca^{2+} from water and Na^+ and K^+ from rocks exchange ions when CAI is negative. In the study region, CAI 1 and CAI 2 values ranged from -2.28 to 0.333 and -0.60 to 0.31 respectively. According to Schoeller classification, CAI 1 and CAI 2 denote 35% of the groundwater samples show a positive ratio denoting the type of direct exchange, and 65% of the groundwater samples indicate a negative ratio denoting reverse exchange (Table 5).

$$\text{CAI 1} = \text{Cl}^- - \text{Na}^+ + \text{K}^+ / \text{Cl}^- \quad (12)$$

$$\text{CAI 2} = \text{Cl}^- - (\text{Na}^+ + \text{K}^+) / \text{SO}_4^{2-} + \text{HCO}_3^- + \text{CO}_3^{2-} + \text{NO}_3^- \quad (13)$$

Table 5

Classification of groundwater based on irrigation water quality parameters in the study region.

Parameter	Water quality	Ranges	Samples numbers	% Of groundwater samples.
SAR (Richards, 1954)	Excellent	0–10	26 (All samples)	100%
	Good	10–18		
	Doubtful	18–26		
EC	Unsuitable	>26	—	
	Excellent	<250		
	Good	250–750		
	Permissible	750–2000		
	Doubtful	2000–3000		
TDS (Freeze and cherry, 1979)	Unsafe	>3000	1	4%
	Fresh	<1000	18	69%
	Brackish	>1000	8	31%
	Saline	>10,000		
	Braine	>100,000		
TH (Sawyer and McCarty, 1967)	Safe	<75		12%
	Moderately hard	75–150		
	Hard	150–300		
RSC (Eaton, 1950a, 1950b)	Very hard	>300	21	88%
	Good	<1.25		
	Doubtful	1.25–2.50		
%Na (Wilcox, 1955)	Unsuitable	>2.50	3	11%
	Excellent	<20	1	4%
	Good	20–40	15	58%
PI (Doneen, 1964)	Permissible	40–60	10	38%
	Doubtful	60–80	—	
	Unsuitable	>80	—	
KR (Kelley, 1940)	unsuitable	<25 class-III	—	
	Good	25–75 class-II	23	88%
	Excellent	>75 class-I	3	12%
MAR (Szabolcs, 1964)	Suitable	<1	23	89%
	Unsuitable	>1	3	11%
CAI-1	Suitable	<50	18	69%
	Unsuitable	>50	8	31%
CAI-2	Positive	Deep meteoric	9	35%
	Negative	Shallow meteoric	17	65%
Soltan (R1)	Positive	Deep meteoric	9	35%
	Negative	Shallow meteoric	17	65%
Soltan (R2)	Positive	Deep meteoric	5	19%
	Negative	Shallow meteoric	21	81%
Soltan (R2)	Positive	Deep meteoric	5	19%
	Negative	Shallow meteoric	21	81%

4.15. Kelley's ratio (KR)

The Kelly's ratio (KR) is the standard measure used to classify irrigation water quality. It determines the concentration of Ca^{2+} and Mg^{2+} ions by analyzing the concentration of Na^+ . The ion concentrations are expressed in meq/L and can be calculated using Eq. (14) to determine the KR.

$$\text{KR} = \frac{\text{Na}^+}{(\text{Ca}^{2+} + \text{Mg}^{2+})} \quad (14)$$

According to Table 4, the KR in the study region ranges from 0.20 to 1.22, with an average of 0.58. Based on KR's classification, about 89% of groundwater samples are suitable for farming, while the remaining 11% are not, as shown in Table 5.

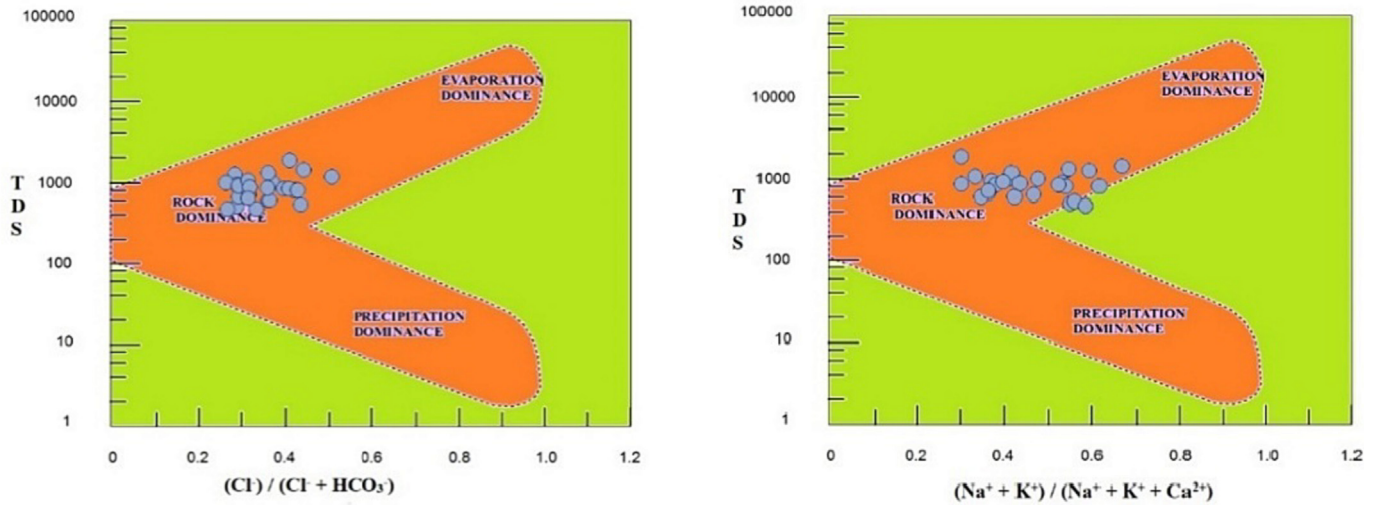


Fig. 7. Plotting in Gibbs diagram for cations and anions vs TDS.

4.16. Soltan classification

According to Soltan's classification, groundwater can be divided into base-exchange indices and meteoric genesis indices. The classification is based on two factors: R1 and R2 (Eq. (15) and Eq. (16)). If the values of both indices are <1 , then the groundwater is of $\text{Na}^+ - \text{SO}_4^{2-}$ and deep meteoric type respectively. Conversely, if both indices are >1 , then the water source is $\text{Na}^+ - \text{HCO}_3^-$ and shallow meteoric type. In the study area, computed values of R1 and R2 ranges from -5.86 to 2.09 with a mean of -0.09 and -5.11 to 2.95 with a mean of 0.09 , respectively. Based on Soltan's classification, 19% of the groundwater samples are of $\text{Na}^+ - \text{SO}_4^{2-}$ and deep meteoric type, while 81% of the samples are of $\text{Na}^+ - \text{HCO}_3^-$ and shallow meteoric type.

$$R1 = (\text{Na}^+ - \text{Cl}^-) / \text{SO}_4^{2-} \quad (15)$$

$$R2 = [(\text{K}^+ + \text{Na}^+) - \text{Cl}^-] / \text{SO}_4^{2-} \quad (16)$$

4.17. Gibbs diagram

According to Gibbs (1970), analyzing the anions $(\text{Cl}^-) / (\text{Cl}^- + \text{HCO}_3^-)$ and cations $(\text{Na}^+ + \text{K}^+) / (\text{Na}^+ + \text{K}^+ + \text{Ca}^{2+})$ of groundwater samples using plots against TDS (Fig. 7) can reveal three types of fields that affect groundwater chemistry: evaporation dominance, precipitation dominance, and rock dominance. Based on the Gibbs diagrams, all samples fall into the rock-dominated zone, indicating that the weathering of rocks with rock-forming minerals primarily controls most aquifers.

5. Conclusion

The current study focuses on the quality of groundwater in parts of the Anantapur district, a region dominated by hard rocks that are crucial for drinking and irrigation. The parameters were compared to WHO, 2012 drinking water acceptability standards, and graphical tools were utilized to assess groundwater quality for irrigation. The following are the findings of the analysis:

The groundwater in the study area is of slightly alkaline quality. The distribution of major ions in the groundwater is found in the following order: $\text{Na}^+ > \text{Ca}^{2+} > \text{Mg}^{2+} > \text{K}^+$ and $\text{HCO}_3^- > \text{SO}_4^{2-} > \text{Cl}^- > \text{NO}_3^-$. The results demonstrate that Ca^{2+} , Mg^{2+} , Na^+ , K^+ , and SO_4^{2-} concentrations in the study region exceed the (WHO, 2012) maximum permissible limit. According to the water quality index, 69%

of the groundwater samples are "very poor," while the remaining 31% are classified as "unfit for drinking purposes due to effective leaching of ions, over-exploitation of groundwater, direct discharge of effluents, and agricultural impact. Water rock interaction and anthropogenic activity play the dominant role for the high concentrations. As a result, suitable treatment and remediation procedures are required prior to human consumption.

Based on Soltan's classification, 19% of the groundwater samples are of $\text{Na}^+ - \text{SO}_4^{2-}$ and deep meteoric type, while 81% of the samples are of $\text{Na}^+ - \text{HCO}_3^-$ and shallow meteoric type. According to the USSL diagram, 77% of groundwater samples fall into the C3S1 category, which denotes water with a high salt concentration and low sodium content, making it ideal for irrigation, while the remaining 15% of groundwater samples fall into the C4S1 category, which denotes water with very saline conditions and low sodium content, making it unsuitable for irrigation.

According to Wilcox classification, 69% of groundwater samples are good to permitted for farming, whereas 27% and 2% are doubtful to unsuitable, and unsuitable, respectively. As per Richards's (1954) classification-based SAR values, in this study site, all groundwater samples (100%) belong to the excellent category for irrigation. According to Eaton (1950a, 1950b) classification of RSC values, 81% of samples fall under the good category for irrigation, 8% and 11% of samples belong to the doubtful and unsuitable category. Regarding Wilcox (1955) classification of %Na values, 4%, 58% and 38% of samples fall under the excellent, good, and for irrigation. According to Szabolcs (1964) classification of MAR values, 69% and 31% of samples fall under the suitable and unsuitable for irrigation. As per Kelley (1940) classification of KR values, 89% and 11% of samples fall under the suitable and unsuitable for irrigation. As per Doneen (1964) classification of PI values, 88% of groundwater samples falls under the good category, while remaining the 11% of samples fall under the excellent for irrigation.

People are advised not to drink the groundwater where the Ca^{2+} , Mg^{2+} , Na^+ , K^+ , and SO_4^{2-} concentration exceed the permissible limits, and have to depend on alternate sources for drinking and domestic use water. To improve groundwater quality, they recommend using appropriate management measures such as limiting living beings, implementing water storage technologies, treating for moisture in the soils, spreading awareness, and establishing a groundwater quality monitoring program. Spatial pattern maps, created with the use of a GIS tool, provided potential data from the entire water quality distribution in the research area and are a useful method for assessment, monitoring, and prospective prediction.

CRediT authorship contribution statement

P. Ravi Kumar: Writing – original draft, Methodology, Conceptualization. **S. Srinivasa Gowd:** Writing – review & editing, Supervision. **C. Krupavathi:** Software, Data curation.

Declaration of competing interest

Authors declare that there is no any conflict of interest.

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References

- Acharya, S., Sharma, S.K., Khandegar, V., 2018. Assessment of groundwater quality by water quality indices for irrigation and drinking in south West Delhi, India. *Data Brief* 18, 2019–2028.
- Adimalla, N., Dhakate, R., Kasarla, A., Taloor, A.K., 2020a. Appraisal of groundwater quality for drinking and irrigation purposes in Central Telangana, India. *Groundw. Sustain. Dev.* 10, 100334.
- Adimalla, N., Marsetty, S.K., Xu, P., 2020b. Assessing groundwater quality and health risks of fluoride pollution in the Shasler Vagu (SV) watershed of Nalgonda, India. *Hum. Ecol. Risk Assess. Int. J.* 26 (6), 1569–1588.
- Agarwal, M., Singh, M., Hussain, J., 2020. Evaluation of groundwater quality for drinking purpose using different water quality indices in parts of Gautam Budh Nagar District, India. *Asian J. Chem.* 32 (5), 1128–1138.
- Aly, A.A., Al-Omran, A.M., Alharby, M.M., 2015. The water quality index and hydrochemical characterization of groundwater resources in Hafar Albatin, Saudi Arabia. *Arab. J. Geosci.* 8 (6), 4177–4190.
- Anusha, B.N., Babu, K.R., Kumar, B.P., Kumar, P.R., Rajasekhar, M., 2022. Geospatial approaches for monitoring and mapping of water resources in semi-arid regions of Southern India. *Environ. Chall.* 8, 100569.
- Badapalli, P.K., Nakkala, A.B., Kottala, R.B., Gugulothu, S., 2022. Geo environmental green growth towards sustainable development in semi-arid regions using physicochemical and geospatial approaches. *Environ. Sci. Pollut. Res.* 1–18.
- Balamurugan, P., Kumar, P.S., Shankar, K., 2020. Dataset on the suitability of groundwater for drinking and irrigation purposes in the Sarabanga River region, Tamil Nadu, India. *Data Brief* 29, 105255.
- Brown, R.M., McClelland, N.L., Deininger, R.A., O'Connor, M.F., 1972. A water quality index –crashing the psychological barrier. *Indicators of Environmental Quality: Proceedings of a Symposium held during the AAAS Meeting in Philadelphia, Pennsylvania, December 26–31, 1971.* Springer US, pp. 173–182.
- Chaurasia, A.K., Pandey, H.K., Tiwari, S.K., Prakash, R., Pandey, P., Ram, A., 2018. Groundwater quality assessment using water quality index (WQI) in parts of Varanasi District, Uttar Pradesh, India. *J. Geol. Soc. India* 92 (1), 76–82.
- Chinthala, K., Somagouni, S.G., Pappaka, R.K., Gudala, H.V., 2023. Ground water quality assessment using water quality index and geographical information system of Mogamururu River basin, YSR Kadapa District, Andhra Pradesh, India. *Emerging Technologies for Water Supply, Conservation and Management.* Springer International Publishing, Cham, pp. 291–313.
- Deepa, S., Venkateswaran, S., 2018. Appraisal of groundwater quality in upper Manimuktha sub-basin, Vellar river, Tamil Nadu, India by using water quality index (WQI) and multivariate statistical techniques. *Mod. Earth Syst. Environ.* 4 (3), 1165–1180.
- Doneen, L.D., 1964. *Notes on Water Quality in Agriculture.* Published as a water sciences and engineering Department of Water Sciences and Engineering, University of California, paper, p. 4001.
- Eaton, F.M., 1950a. Significance of carbonates in irrigation waters. *Soil Sci.* 69 (2), 123,134.
- Eaton, F.M., 1950b. Significance of carbonates in irrigation waters. *Soil Sci.* 69 (2), 123–134.
- Etikala, B., Golla, V., Adimalla, N., Marapatla, S., 2019. Factors controlling groundwater chemistry of Renigunta area, Chittoor District, Andhra Pradesh, South India: a multivariate statistical approach. *HydroResearch* 1, 57–62.
- Etikala, B., Adimalla, N., Madhav, S., Somagouni, S.G., Keshava Kiran Kumar, P.L., 2021. Salinity problems in groundwater and management strategies in arid and semi-arid regions. *Groundw. Geochem.* 42–56.
- Freeze, R.A., Cherry, J.A., 1979. *Groundwater.* Prentice-Hall Inc, Eaglewood Cliffs, NJ.
- Gibbs, R.J., 1970. Mechanisms controlling world water chemistry. *Science* 170 (3962), 1088–1090.
- Giri, A., Bharti, V.K., Kalia, S., Kumar, K., Khansu, M., 2022. Hydrochemical and quality assessment of irrigation water at the trans-himalayan high-altitude regions of Leh, Ladakh, India. *Appl Water Sci* 12 (8), 197.
- Gnanachandrasamy, G., Ramkumar, T., Venkatramanan, S., Vasudevan, S., Chung, S.Y., Bagyaraj, M., 2015. Accessing groundwater quality in lower part of Nagapattinam district, Southern India: using hydrogeochemistry and GIS interpolation techniques. *Appl Water Sci* 5 (1), 39–55.
- Golla, V., Badapalli, P.K., Mannala, P., 2021. Assessment of groundwater quality for drinking and irrigation in semi-arid regions of Andhra Pradesh, Southern India, using multivariate statistical analysis. *Arab. J. Geosci.* 14 (19), 1–11.
- Gowd, S.S., 2005. Assessment of groundwater quality for drinking and irrigation purposes: a case study of Peddavanka watershed, Anantapur District, Andhra Pradesh, India. *Environ. Geol.* 48 (6), 702–712.
- Jatwa, P., Shrivastava, S., Bhasin, S., Shukla, A.N., 2019. Water quality monitoring of three Lentic water bodies of Ujjain District (MP) India. *Environ. Conserv. J.* 20 (1&2), 101–108.
- Kelley, W.P., 1940. Permissible composition and concentration of irrigation water. *Proceedings of the American Society of Civil Engineers.* vol. 66, pp. 607–613.
- Khatir, N., Tyagi, S., Rawtani, D., Tharmavaram, M., Kamboj, R.D., 2020. Analysis and assessment of ground water quality in Satlasana Taluka, Mehsana district, Gujarat, India through application of water quality indices. *Groundw. Sustain. Dev.* 10, 100321.
- Krishan, G., Singh, S., Kumar, C.P., Garg, P., Suman, G., Ghosh, N.C., Chaudhary, A., 2016. Assessment of groundwater quality for drinking purpose by using water quality index (WQI) in Muzaffarnagar and Shamli districts, Uttar Pradesh, India. *Hydrol. Curr. Res.* 7 (227), 2.
- Krishnakumar, P., Lakshumanan, C., Kishore, V.P., Sundararajan, M., Santhiya, G., Chidambaram, S., 2014. Assessment of groundwater quality in and around Vedaraniyam, South India. *Environ. Earth Sci.* 71 (5), 2211–2225.
- Krupavathi, C., Gowd, S.S., Kumar, P.R., Vijay, G.H., Kumar, B.P., 2022. Spatial analysis of groundwater qualities in Vempalle Mandal of YSR District, Andhra Pradesh, India using geospatial techniques. *Hydrospat. Anal.* 6 (1), 1–12.
- Kumar, P.R., Gowd, S.S., Kumar, B.P., Krupavathi, C., 2021. Hydrogeochemical studies of Kanaganapalle Mandal, Anantapur District, Andhra Pradesh by using geospatial techniques. *Kala Sarovar J.* 24 (3), 54–65.
- Kumar, P.R., Gowd, S.S., Krupavathi, C., Kumar, B.P., 2022a. Groundwater quality assessment using water quality index (WQI) in parts of Anantapur, A.P. (India). *J. Ind. Geophys. Union* 26 (2), 155–164.
- Kumar, P.R., Gowd, S.S., Krupavathi, C., Vijay, G.H., 2022b. Assessment of Groundwater quality using water quality index (WQI) and pollution index of Groundwater (PIG) and GIS parts of Anantapur District, Andhra Pradesh, India. *J. Remote Sens. GIS* 13 (3), 1–14 ISSN-2230-7990.
- Kumar, B.P., Anusha, B.N., Babu, K.R., Sree, P.P., 2023. Identification of climate change impact and thermal comfort zones in semi-arid regions of AP, India using LST and NDBI techniques. *J. Clean. Prod.* 407, 137175.
- Kumari, M., Rai, S.C., 2020. Hydrogeochemical evaluation of groundwater quality for drinking and irrigation purposes using water quality index in semi-arid region of India. *J. Geol. Soc. India* 95 (2), 159–168.
- Lakshmi, R.V., Raja, V., Sekar, C.P., Neelakantan, M.A., 2021. Evaluation of groundwater quality in Virudhunagar Taluk, Tamil Nadu, India by using statistical methods and GIS technique. *J. Geol. Soc. India* 97 (5), 527–538.
- Mahmud, A., Sikder, S., Joardar, J.C., 2020. Assessment of groundwater quality in Khulna city of Bangladesh in terms of water quality index for drinking purpose. *Appl Water Sci* 10, 1–14.
- Pande, C.B., Moharir, K.N., Singh, S.K., Dzwairo, B., 2020. Groundwater evaluation for drinking purposes using statistical index: study of Akola and Buldhana districts of Maharashtra, India. *Environ. Dev. Sustain.* 22 (8), 7453–7471.
- Papazotos, P., Koumantakis, I., Vasileiou, E., 2019. Hydrogeochemical assessment and suitability of groundwater in a typical Mediterranean coastal area: a case study of the Marathon basin, NE Attica, Greece. *HydroResearch* 2, 49–59.
- Pasham, H., Gugulothu, S., Badapalli, P.K., Dhakate, R., Kottala, R.B., 2022. Geospatial approaches of TGIS and morphometric analysis in the Mahi River basin using Landsat 8 OLI/TIRS and SRTM-DEM. *Environ. Sci. Pollut. Res.* 1–18.
- Prasad, M., Sunitha, V., Reddy, Y.S., Suvama, B., Reddy, B.M., Reddy, M.R., 2019. Data on water quality index development for groundwater quality assessment from Obulavaripalli Mandal, YSR district, AP India. *Data Brief* 24, 103846.
- Ram, A., Tiwari, S.K., Pandey, H.K., Chaurasia, A.K., Singh, S., Singh, Y.V., 2021. Groundwater quality assessment using water quality index (WQI) under GIS framework. *Appl Water Sci* 11 (2), 1–20.
- Ravindra, B., Subba Rao, N., Dhananjaya Rao, E.N., 2023. Groundwater quality monitoring for assessment of pollution levels and potability using WPI and WQI methods from a part of Guntur district, Andhra Pradesh, India. *Environ. Dev. Sustain.* 25 (12), 14785–14815.
- Rezaei, A., Hassani, H., Tziritis, E., Mousavi, S.B.F., Jabbari, N., 2020. Hydrochemical characterization and evaluation of groundwater quality in Dalgan basin, SE Iran. *Groundw. Sustain. Dev.* 10, 100353.
- Richards, L.A., 1954. *Diagnosis and Improvement of Saline and Alkali Soils.* Handbook, p. 60.
- Safari, M., Ahmadvazeli, A., Ghanbari, A., Mokhtari, Z., Soleimani, Z., 2021. Assessment of the HablehRood river water quality for drinking and irrigation purposes in Garmsar, Iran. *Environ. Earth Sci.* 80, 1–11.
- Samtio, M.S., Rajper, K.H., Mastoi, A.S., Sadaf, R., Rajper, R.H., Hakro, A.A., ... Lanjwani, M.F., 2023. Hydrochemical assessment of groundwater from taluka Dahili, Thar Desert, Pakistan, for irrigation purpose using water quality indices. *Int. J. Environ. Anal. Chem.* 103 (10), 2368–2384.
- Sarkar, M., Pal, S.C., Islam, A.R.M., 2022. Groundwater quality assessment for safe drinking water and irrigation purposes in Malda district, Eastern India. *Environ. Earth Sci.* 81 (2), 1–20.
- Sawyer, C.N., McCarty, P.L., 1967. *Chemistry for Sanitary Engineers.*
- Schoeller, H., 1977. *Geochemistry of groundwater.* Groundwater Studies. An International Guide for Research and Practice. 15. UNESCO, Paris, pp. 1–18.

- Singha, S.S., Pasupuleti, S., 2020. Hydrogeochemical modeling-based approach for evaluation of groundwater suitability for irrigational use in Korba district, Chhattisgarh, Central India. *SN Appl. Sci.* 2 (9), 1–13.
- Subba Rao, N., 2018. Groundwater quality from a part of Prakasam district, Andhra Pradesh, India. *Appl Water Sci* 8 (1), 1–18.
- Swarna Latha, P., Nageswara Rao, K., 2012. An integrated approach to assess the quality of groundwater in a coastal aquifer of Andhra Pradesh, India. *Environ. Earth Sci.* 66 (8), 2143–2169.
- Szabolcs, I., 1964. The influence of irrigation water of high sodium carbonate content on soils. *Agrokém. Talajt.* 13 (sup), 237–246.
- Tiwari, A.K., Singh, A.K., Singh, A.K., Singh, M.P., 2017. Hydrogeochemical analysis and evaluation of surface water quality of Pratapgarh district, Uttar Pradesh, India. *Appl Water Sci* 7 (4), 1609–1623.
- Todd, D.K., Mays, L.W., 2004. *Groundwater Hydrology*. John Wiley & Sons.
- USSR, 1954. *Diagnosis and improvement of salinity and alkaline soil*. USDA Hand Book no. 60, Washington.
- Wang, S., 2013. Groundwater quality and its suitability for drinking and agricultural use in the Yanqi Basin of Xinjiang Province, Northwest China. *Environ. Monit. Assess.* 185 (9), 7469–7484.
- Wilcox, L.V., 1955. *Classification and Use of Irrigation Water*. USDA circular 969, 19 p. Washington, DC USA.
- World Health Organization, 2012. *Good Health Adds Life to Years: Global Brief for World Health Day 2012* (No. WHO/DCO/WHO/2012.2). World Health Organization.